

Fast Electromagnetic Transient Modeling of Transformers Based on the Partial Element Equivalent Circuit and Image Methods

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Abstract—Accurate and physically correct white-box transient modeling of transformers is essential for both fault diagnosis and design. Conventional finite element method (FEM)-based approaches, although widely adopted, are computationally intensive and often necessitate simplifications of transformer windings. In such models, turns or disks are typically treated as isolated conductors without connections, which hinders the construction of practical winding configurations. However, precise winding modeling is critical for capturing the fast transient behavior of transformers. To address these challenges, this study proposes a transient modeling framework for transformers based on the partial element equivalent circuit (PEEC) and image methods. In this framework, the equivalent inductance, capacitance, and resistance matrices of the winding are derived using the PEEC method, and subsequently adjusted through the image method to account for the boundary effects of the core and tank. Finally, a coupled circuit-magnetic circuit model is proposed to consider the core flux. The effectiveness of the proposed method is verified through FEM-based simulations and two practical cases. The simulation results show that the maximum deviation among the matrix elements remains within 9%, while the computational speed improves by a factor of approximately 2000. Furthermore, the experimental results confirm that the proposed method can accurately capture the fast transient behavior of transformers.

Index Terms—Boundary effects, magnetic circuit, PEEC, transformer modeling, transient simulation, image method.

I. INTRODUCTION

TRANSFORMERS, often regarded as the heart of power systems, are essential for voltage conversion and stable energy transmission, with their reliability directly impacting the grid stability [1], [2]. Numerous studies [3]–[5] have highlighted the catastrophic risks associated with transformer failures, which can trigger widespread blackouts and result

in economic losses. Approximately 70% of these failures are attributed to transient overvoltages and insulation breakdown, both closely related to electromagnetic transients [6], [7]. Therefore, accurate transient simulation is essential for transformers, offering crucial guidance for design, operation, and maintenance to prevent catastrophic failures [8]–[11].

Three types of transformer modeling approaches are widely used for fast transient simulations in both time and frequency domains: black-box, gray-box, and white-box modeling [2]. Each method exhibits distinct characteristics, offering specific advantages as well as inherent limitations.

Black-box modeling represents the transformer as a system approximated by cascaded RLC units or by neural networks that directly fit terminal responses [12]–[14]. These approaches are computationally efficient and capable of reproducing measured responses without requiring detailed knowledge of the physical geometry. However, due to their lack of physical interpretability, black-box models are mainly confined to system-level simulations where only terminal responses are required. Building on black-box concepts, gray-box modeling employs a physically informed equivalent circuit topology, with parameters such as inductance, capacitance, and resistance identified through optimization algorithms including genetic algorithms, particle swarm optimization, and Bayesian optimization [2], [15]–[19]. This strategy enhances fitting accuracy while preserving limited physical interpretability. However, parameter identification typically requires extensive search-space exploration and iterative tuning, which is computationally costly and remains only partially interpretable. In addition, most studies restrict their focus to single windings, neglecting coupling between high-voltage (HV) and low-voltage (LV) windings or among different phases. Taken together, black-box and gray-box models provide efficient and reasonably accurate representations of transformer behavior, particularly for system-level studies. Yet, their limited physical interpretability reduces their effectiveness in guiding transformer design and fault analysis, thereby motivating the development of white-box approaches.

White-box models are derived directly from the physical laws, geometrical structures, and material properties of transformers. They explicitly capture the device's internal electromagnetic behavior, thereby providing physically interpretable representations of transformer characteristics. Among them, finite-element method (FEM)-based approaches are widely used as rigorous field-solver references for extracting inductance, capacitance, and resistance matrices [2], [20], [21]. Owing to their capability to represent complex geometries and material properties, FEM provides an important benchmark for validating other modeling methods [22]. How-

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ever, when 3D FEM is employed for transformer modeling, full-domain meshing and field solutions are required. Consequently, the computational burden becomes particularly significant in parametric studies, insulation-design iterations, and optimization [2], where winding configurations or design parameters must be adjusted iteratively. Moreover, practical winding connections are not always straightforward to incorporate into FEM-based transformer models. In many cases, windings across different disks or turns are treated as independent conductors to obtain the required loop currents or potential differences for parameter extraction [2], [23]. As a result, practical connections among turns, disks, taps, and interleaved sections are often difficult to represent. An alternative white-box approach is to directly calculate the inductance, capacitance, and resistance matrices using physical formulas [8], [24]. While this approach significantly reduces computation time, it applies only to the winding itself, neglecting other transformer components, and cannot adequately represent complex winding configurations.

From an engineering application perspective, white-box models remain essential for transformer manufacturers and researchers because they retain the link between physical structure and internal electrical quantities. In this context, the partial element equivalent circuit (PEEC) method provides an efficient solution to FEM. Derived from the integral-equation formulation of Maxwell's equations [25], PEEC mainly discretizes conductors rather than the entire surrounding field domain, thereby reducing the computational burden for modeling transformers [22], [26], [27]. Moreover, the partial formulation of PEEC enables discretization along the conductor path, making practical winding connections straightforward to represent and providing access to internal voltage distributions at any discretized segment. These two features are particularly useful for fast transient studies. However, directly transferring mature PEEC formulations to transformer modeling is not feasible, as transformers exhibit unique structural and electromagnetic characteristics that necessitate specific adaptations. In particular, conventional PEEC implementations rarely account for boundary effects introduced by the transformer core and tank. These limitations hinder the applicability of PEEC in transformer transient studies.

To overcome these challenges, this study introduces a fast electromagnetic transient modeling framework for transformers based on the PEEC and image methods. The main contributions are as follows:

- 1) We propose a PEEC-based transient modeling framework for transformers. This approach complements existing FEM-based white-box models by facilitating the representation of complex winding configurations and enabling computationally efficient transient simulations.
- 2) We integrate the PEEC method with the image method to capture the boundary effects introduced by the transformer core and tank at high frequencies. The results demonstrate that the proposed approach is simple, effective, and capable of satisfying accuracy requirements.
- 3) We develop a coupled circuit-magnetic circuit model that explicitly distinguishes between flux inside and outside the core, thereby compensating for the image-

method assumption that the transformer core behaves as a magnetic barrier.

The remainder of this study is organized as follows. Section II presents the proposed methodology. Section III employs FEM-based simulations to verify the effectiveness of the method, while Section IV validates it through two practical cases. Section V discusses several issues related to this study, and Section VI concludes this study.

II. METHODOLOGY

As shown in Fig. 1, this section first presents transformer winding modeling using the PEEC method, then adjusts the PEEC-derived matrices by incorporating core and tank boundary effects through the image method, and finally considers the influence of core flux.

A. PEEC-Based Modeling of Transformer Windings

The PEEC method is derived from the integral formulation of Maxwell's equations and provides a systematic way to represent electromagnetic interactions in terms of equivalent circuit parameters [25]–[28]. For a conductor segment, the electric field can be expressed as:

$$\mathbf{E}(\mathbf{r}) = \frac{\mathbf{J}(\mathbf{r})}{\sigma} + j\omega \mathbf{A}(\mathbf{r}) + \nabla\phi(\mathbf{r}) \quad (1)$$

where $\mathbf{J}$ is the current density, σ is the conductivity, $\mathbf{A}$ is the magnetic vector potential, ϕ is the electric scalar potential, $\mathbf{r}$ denotes the position vector of the field point (referred to as the field segment in this study), and ω is the angular frequency. The potentials $\mathbf{A}$ and ϕ are obtained through the Green's function, enabling the derivation of equivalent circuit parameters that characterize inductive and capacitive couplings. Accordingly, the electric field at location $\mathbf{r}$ can be expressed in terms of the Green's function as:

$$\mathbf{E}(\mathbf{r}) = \frac{\mathbf{J}(\mathbf{r})}{\sigma} + j\omega\mu \int_{V'} G(\mathbf{r}, \mathbf{r}') \mathbf{J}(\mathbf{r}') dV' + \frac{\nabla}{\varepsilon} \int_{S'} G(\mathbf{r}, \mathbf{r}') q(\mathbf{r}') dS' \quad (2)$$

where μ is the magnetic permeability of the medium, ε is the dielectric permittivity of the medium, $\mathbf{r}'$ denotes the position vector of the source point (referred to as the source segment in this study), and $G(\mathbf{r}, \mathbf{r}')$ is the Green's function characterizing the spatial propagation of electromagnetic interactions, which in homogeneous non-conductive medium is expressed as:

$$G(\mathbf{r}, \mathbf{r}') = \frac{\exp(j\omega\sqrt{\mu\varepsilon}|\mathbf{r} - \mathbf{r}'|)}{4\pi|\mathbf{r} - \mathbf{r}'|} \quad (3)$$

To solve Eq. (2), the continuous current and charge distributions must be discretized into finite segments. In the PEEC method, conductors are discretized into volume cells for current density and surface cells for charge density. This discretization transforms the integral equations into summations over discrete segments, thereby enabling the extraction of equivalent circuit parameters.

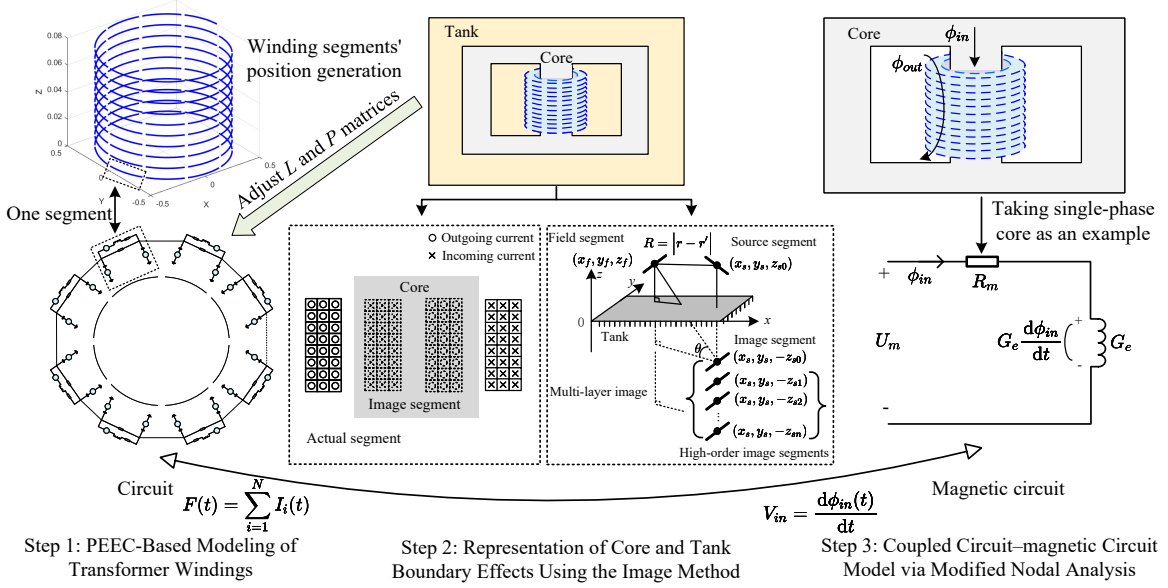


Fig. 1. Overall framework of the proposed electromagnetic transient model for transformers. This figure summarizes the main modeling procedure, including winding discretization, PEEC matrix extraction, image-method-based treatment of boundary effects, and coupled circuit-magnetic circuit formulation via modified nodal analysis.

For the transformer, the windings are discretized into finite segments, as shown in Fig. 1. The partial inductance between two segments m and n is defined as:

$$L_{mn} = \frac{\mu}{4\pi} \int_{V_m} \int_{V_n} \frac{\mathbf{u}_m \cdot \mathbf{u}_n}{|\mathbf{r}_m - \mathbf{r}_n|} dV_m dV_n \quad (4)$$

where $\mathbf{u}_m$ and $\mathbf{u}_n$ are unit vectors along the current directions, and $|\mathbf{r}_m - \mathbf{r}_n|$ is the distance between the two segments. Similarly, the potential coefficient is given by:

$$P_{mn} = \frac{1}{4\pi\epsilon S_m S_n} \int_{S_m} \int_{S_n} \frac{1}{|\mathbf{r}_m - \mathbf{r}_n|} dS_m dS_n \quad (5)$$

In addition, the resistance of each winding segment is calculated using the resistivity formula:

$$R_m = \frac{l_m}{\sigma S_m} \quad (6)$$

where l_m is the length of each segment, and S_m is the cross-sectional area of each segment. Through this discrete computational process, the integral formulation of Maxwell's equations is mapped into equivalent inductance $\mathbf{L}$, potential coefficient $\mathbf{P}$ (with capacitance $\mathbf{C} = \mathbf{P}^{-1}$), and resistance $\mathbf{R}$ matrices, which constitute the basis of the equivalent circuit for transformer windings, as shown in Fig. 2.

In the proposed PEEC-based framework, the winding is discretized along the conductor direction, and each segment is associated with the corresponding entries in the partial inductance $\mathbf{L}$, potential coefficient $\mathbf{P}$, and resistance $\mathbf{R}$ matrices. The discretization points should be arranged according to the practical winding connection topology, such as turns, disks, layers, tapping positions, and inter-disk connections. In addition, the extracted segment-level parameters can be assembled into equivalent circuits at various modeling scales (e.g., turn-, disk-, or winding-level models) through node-merging operations [22]. It should be noted that the conductor

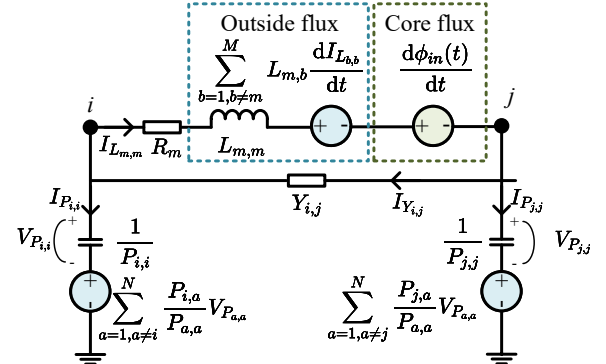


Fig. 2. Equivalent circuit of each segment using the PEEC method. The outside flux represented by the $\mathbf{L}$ matrix denotes flux outside the core, while the core flux corresponds to flux inside the core. By removing the core flux, the circuit reduces to a standard PEEC representation. $Y_{i,j}$ typically represents the leakage current between adjacent segments. Since the transformer windings are wrapped in insulating paper, $Y_{i,j}$ is set to 0 in this study.

cross-section is not further discretized in this study. Each winding segment is modeled with a uniformly distributed equivalent current over its cross section, without accounting for current redistribution caused by skin and proximity effects.

B. Representation of Core and Tank Boundary Effects Using the Image Method

The parameters L_{mn} and P_{mn} derived in the previous section are referenced to infinity [25], where the magnetic vector potential and the electric potential are assumed to vanish. However, in practical transformers exposed to high-frequency transients, the presence of cores and tanks introduces boundary conditions analogous to those in FEM, thereby modifying the electromagnetic field distribution through changes in inductive and capacitive couplings among winding segments [24]. To

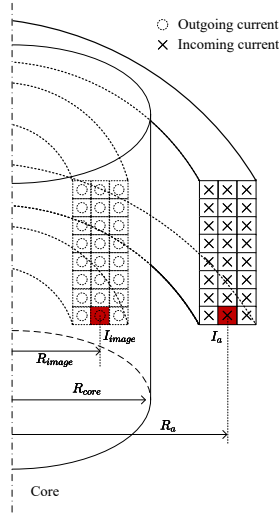


Fig. 3. Schematic illustration of image segments introduced to represent actual segments under the boundary effect of the transformer core.

properly account for these boundary effects, the calculated results must be adjusted. Accordingly, this section incorporates the boundary effects of the core and tank using the image method, in which both are approximated as ideal conductors with zero electric potential and magnetic insulation.

The transformer core can be idealized as an infinitely long coaxial cylindrical surface. However, obtaining a closed-form analytical solution for the image segment of the winding with respect to such a cylindrical boundary is not straightforward. To address this difficulty, an approximation is adopted by considering the image segment of the winding relative to a sphere, which is validated by subsequent results as sufficiently accurate. For an actual segment (i.e., winding segment) carrying current I_a , an image segment carrying current I_{image} is introduced to represent the boundary effect of the core, as illustrated in Fig. 3. The geometric and current relationships between the actual segment and its image segment are expressed as:

$$R_{image} = \frac{R_{core}^2}{R_a} \quad (7)$$

$$I_{image} = -\frac{R_a}{R_{core}} I_a \quad (8)$$

where R_{core} is the radius of the transformer core and R_a is the radius of the actual segment. Therefore, all image segments corresponding to discretized transformer winding segments can be obtained from Eq. (7). By subsequently calculating the $\mathbf{L}_{core}$ and $\mathbf{P}_{core}$ matrices of these image segments using Eqs. (4)-(5), the boundary effect of the transformer core is incorporated into $\mathbf{L}$ and $\mathbf{P}$ as follows:

$$\mathbf{L} = \mathbf{L} - \frac{R_a}{R_{core}} \mathbf{L}_{core} \quad (9)$$

$$\mathbf{P} = \mathbf{P} - \frac{R_a}{R_{core}} \mathbf{P}_{core} \quad (10)$$

Regarding the boundary effect of the transformer tank, the cuboid tank is approximated using the first-order image method [24]. For each winding segment, 26 image segments

are introduced, including 6 face image segments from single reflections, 12 edge image segments from two successive reflections, and 8 corner image segments from three successive reflections, as shown in Fig. 4. The current direction and charge polarity depend on the number of reflections: odd-order reflections, corresponding to the face and corner image segments, reverse the current direction and charge sign, whereas even-order reflections, corresponding to the edge image segments, preserve them. All image segments retain the same current and charge magnitudes as the original segment. In this study, only the first-order image method is used to maintain computational efficiency.

The boundary effect of the transformer tank is incorporated into $\mathbf{L}$ and $\mathbf{P}$ as follows:

$$L_{mm} = L_{mm} + \sum_i^{12} L_{mi}^+ - \sum_i^{14} L_{mi}^- \quad (11)$$

$$P_{mm} = P_{mm} + \sum_i^{12} P_{mi}^+ - \sum_i^{14} P_{mi}^- \quad (12)$$

where L_{mm} and P_{mm} are the self-inductance and self-potential coefficient of the actual segment m , respectively. Here, L_{mi}^+ and P_{mi}^+ correspond to the mutual inductance and potential coefficient, respectively, between the actual segment m and the positive image segment i , while L_{mi}^- and P_{mi}^- are defined similarly for the negative image segments. The positions of the image segments are illustrated in Fig. 4a and Fig. 4c.

$$L_{mn} = L_{mn} + \sum_i^{12} L_{mi}^+ - \sum_i^{14} L_{mi}^- \quad (13)$$

$$P_{mn} = P_{mn} + \sum_i^{12} P_{mi}^+ - \sum_i^{14} P_{mi}^- \quad (14)$$

where L_{mn} and P_{mn} denote the mutual inductance and mutual potential coefficient between the field segment m and the source segment n , respectively. L_{mi}^+ and L_{mi}^- represent mutual inductance between the field segment m and the positive and negative image segment i of source segment n , while P_{mi}^+ and P_{mi}^- denote the corresponding mutual potential coefficients. The spatial arrangements of these image segments are illustrated in Fig. 4b and Fig. 4d.

C. Inclusion of Core Flux via Coupled Circuit-Magnetic Circuit Model

At high frequencies, the eddy current effect significantly suppresses magnetic flux penetration into the transformer core [29]. Hence, the core flux can be reasonably assumed to be nearly zero, and the image method enables accurate simulation of the core boundary effect. However, at low frequencies, magnetic fields may penetrate the core, rendering the impact of core flux non-negligible. To account for this effect, a coupled circuit-magnetic circuit model is introduced to improve transient-voltage calculation. Specifically, the standard PEEC-derived equivalent circuit (Fig. 2) is extended by incorporating the core flux into the modified nodal analysis

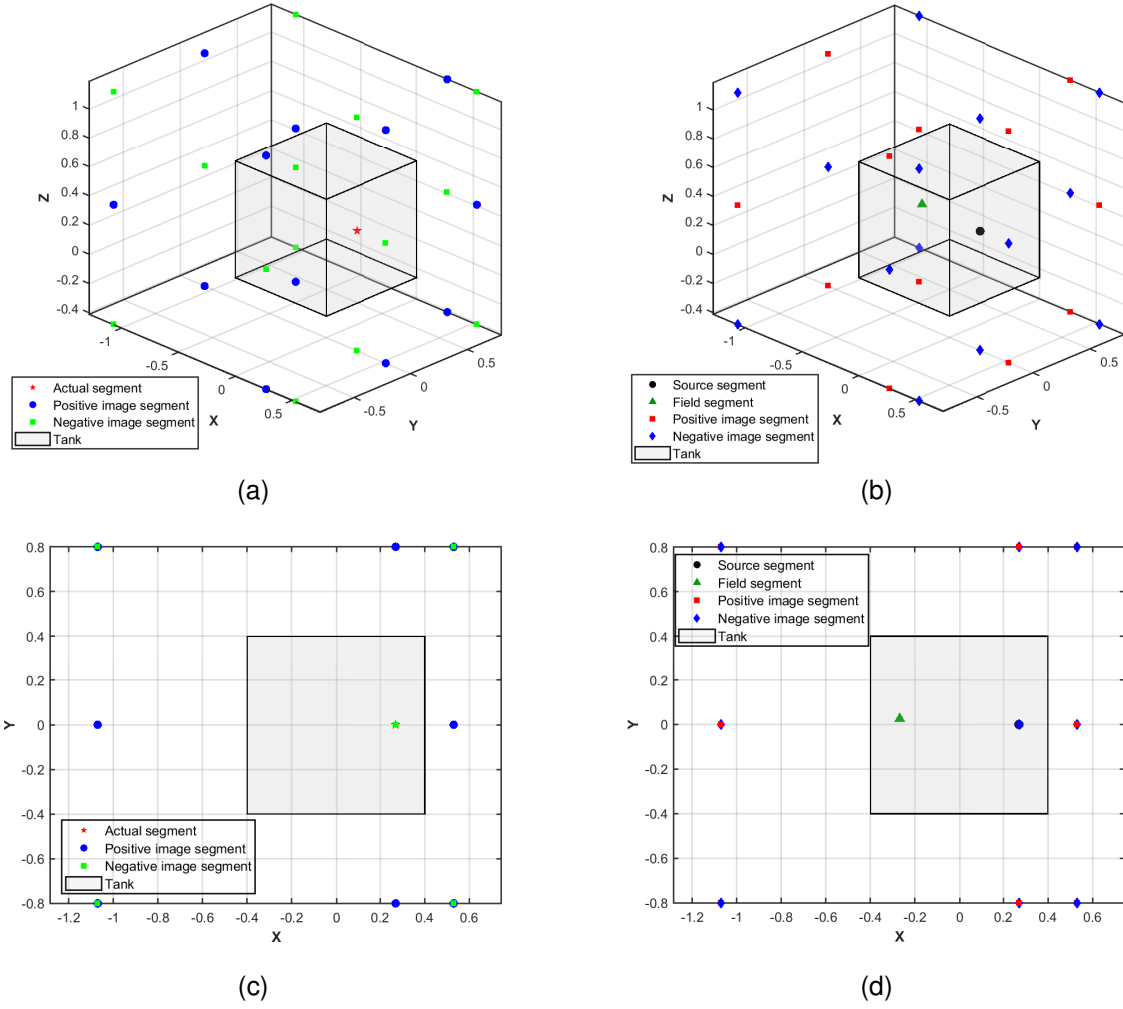


Fig. 4. Schematic illustration of image segments introduced to represent actual segments under the boundary effect of the transformer tank ($0.8m \times 0.8m \times 0.8m$). The positions of these image segments are symmetric with respect to the tank walls. (a) Three-dimensional schematic of image segments used in the calculation of L_{mm} and P_{mm} . (b) Three-dimensional schematic of image segments used in the calculation of L_{mn} and P_{mn} ($m \neq n$). (c) Projection of (a) onto the XOY plane. (d) Projection of (b) onto the XOY plane.

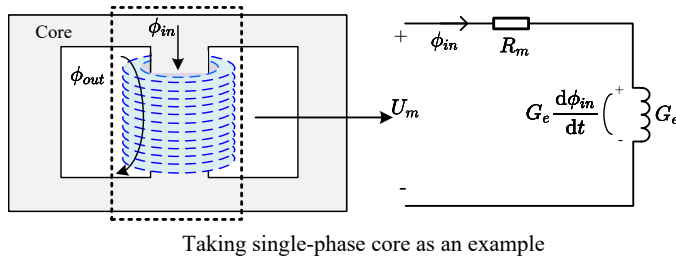


Fig. 5. Magnetic circuit representation of a single-phase transformer core.

(MNA) formulation. The induced voltage resulting from variations in the core flux is introduced as a compensation term, analogous to the treatment of lossy ground in transmission line theory [30].

$$\begin{bmatrix} -\mathbf{A}_{circuit} & -(\mathbf{R} + \mathbf{L} \frac{d}{dt}) \\ \mathbf{G} + \mathbf{P}^{-1} \frac{d}{dt} & -\mathbf{A}_{circuit}^T \end{bmatrix} \begin{bmatrix} \mathbf{V}_n(t) \\ \mathbf{I}_b(t) \end{bmatrix} = \begin{bmatrix} \mathbf{V}_s(t) + \frac{d\phi_{in}(t)}{dt} \\ \mathbf{I}_s(t) \end{bmatrix} \quad (15)$$

where $\mathbf{A}_{circuit}$ denotes the incidence matrix; $\mathbf{R}$, $\mathbf{L}$, $\mathbf{P}$, and $\mathbf{G}$ represent the resistance, inductance, potential coefficient, and conductance matrices, respectively. $\mathbf{V}_n(t)$ is the nodal voltage vector, $\mathbf{I}_b(t)$ is the branch current vector, while $\mathbf{V}_s(t)$ and $\mathbf{I}_s(t)$ denote the applied branch voltage source and nodal current source, respectively. $\phi_{in}(t)$ represents the core flux. When the term $\frac{d\phi_{in}(t)}{dt}$ is neglected, Eq. (15) reduces to the basic MNA formulation.

As shown in Fig. 5, the total magnetic flux of a single-phase winding comprises two components: outside flux ϕ_{out} , located between the core and the winding, and the core flux ϕ_{in} , confined within the core. Before calculating the core flux, the contribution of the outside flux should be subtracted from the magnetomotive force. For a single-phase transformer core, the core flux can be determined using the following formulas:

$$U_m(t) = \sum_{b=1}^{N_b} I_b(t) - R_{out} \phi_{out}(t) \quad (16)$$

$$\phi_{out}(t) = \mathbf{1}_{N_b \times 1} \mathbf{L} \mathbf{I}_b(t) \quad (17)$$

$$U_m(t) = R_m \phi_{in}(t) + G_e \frac{d\phi_{in}(t)}{dt} \quad (18)$$

where $U_m(t)$ denotes the magnetomotive force of the core magnetic circuit; I_b represents the branch current at the turn-level modeling scale; N_b is the number of winding turns; R_{out} is the reluctance outside the core, calculated according to Ref. [31], [32]; R_m corresponds to the core reluctance, determined following Ref. [33], [34]; and G_e denotes the equivalent eddy-current inductance in the magnetic circuit model of the transformer core, characterizing the damping effect of eddy currents on rapid core flux variations, as described in Ref. [33], [34]. The definitions of R_m and G_e are provided in the Appendix. With these parameters, the coupled circuit-magnetic circuit model behaves as a low-pass filter. The corresponding cutoff frequency can be estimated by $f_c = R_m / (2\pi G_e)$, implying that the influence of the core flux becomes negligible at high frequencies, as reported in existing Refs. [2], [15]–[17], [20], [23].

Algorithm 1 Time-Domain Solver with Coupled Circuit–Magnetic Circuit Model

Input: Incidence matrix $\mathbf{A}_{circuit}$, resistance matrix $\mathbf{R}$, inductance matrix $\mathbf{L}$, admittance matrix $\mathbf{G}$, voltage sources $\mathbf{V}_s(t)$, current sources $\mathbf{I}_s(t)$, time step dt , number of time steps N_t , core reluctance R_m , equivalent eddy-current inductance G_e , and leakage reluctance $R_{leakage}$

Output: Nodal voltages $\mathbf{V}_n(t)$ and branch currents $\mathbf{I}_b(t)$

- 1: Determine branch and nodal counts: $[N_b, N_n] \leftarrow \text{size}(\mathbf{A}_{circuit})$
 - 2: Allocate $\mathbf{I}_b(t) \leftarrow \mathbf{0}^{N_b \times N_t}$, $\mathbf{V}_n(t) \leftarrow \mathbf{0}^{N_n \times N_t}$, $\Phi_{in} \leftarrow \mathbf{0}^{N_b \times N_t}$, $\phi_{in} \leftarrow \mathbf{0}^{N_t}$
 - 3: Construct matrices:
 $\mathbf{Z}_{left} = \mathbf{R} + \mathbf{L}/dt$
 $\mathbf{C}_{left} = \mathbf{G} + \mathbf{P}^{-1}/dt$
 $\mathbf{LEFT} = \begin{bmatrix} -\mathbf{A}_{circuit} & -\mathbf{Z}_{left} \\ \mathbf{C}_{left} & -\mathbf{A}_{circuit}^T \end{bmatrix}$
 - 4: **for** $k_t = 2$ to N_t **do**
 - 5: Retrieve excitations:
 $\mathbf{v}_s = \mathbf{V}_s(:, k_t)$
 $\mathbf{i}_s = \mathbf{I}_s(:, k_t)$
 - 6: Assemble right-hand side vector:
 $\mathbf{V}_{in} = (\Phi_{in}(:, k_t) - \Phi_{in}(:, k_t - 1))/dt$
 $\mathbf{RIGHT} = \begin{bmatrix} \mathbf{v}_s - (\mathbf{L}/dt)\mathbf{I}_b(:, k_t - 1) + \mathbf{V}_{in} \\ \mathbf{i}_s + \mathbf{C}_{left}\mathbf{V}_n(:, k_t - 1) \end{bmatrix}$
 - 7: Solve:
 $\mathbf{x} = \mathbf{LEFT}^{-1}\mathbf{RIGHT}$
 - 8: Extract:
 $\mathbf{V}_n(:, k_t) = \mathbf{x}(1 : N_n)$
 $\mathbf{I}_b(:, k_t) = \mathbf{x}(N_n + 1 : \text{end})$
 - 9: Compute magnetomotive force:
 $\phi_{out}(k_t) = \mathbf{1}_{N_b \times 1} \mathbf{L} \mathbf{I}_b(k_t)$
 $U_m(k_t) = \sum_{b=1}^{N_b} I_b(k_t) - R_{out} \phi_{out}(k_t)$
 - 10: Update core flux:
 $\phi_{in}(k_t + 1) = \frac{[U_m(k_t) - R_m \phi_{in}(k_t)]dt}{G_e} + \phi_{in}(k_t)$
 $\Phi_{in}(k_t + 1) = \phi_{in}(k_t + 1) \mathbf{1}_{N_b \times 1}$
 - 11: **end for**
-

model, and its time-domain solution procedure is outlined in Algorithm 1. The backward Euler method is employed to calculate nodal voltages $\mathbf{V}_n(t)$ and branch currents $\mathbf{I}_b(t)$, while the forward Euler method is applied to calculate the core flux $\phi_{in}(t)$ [35]. Moreover, since the magnetomotive force must be evaluated at the turn level in Eq. (16), the $\mathbf{L}$, $\mathbf{P}$, and $\mathbf{R}$ matrices are required to be computed accordingly at the turn level. In contrast, when the effect of the core flux is neglected, matrices at any discretization level can be employed in the solution. As an illustrative case, this study presents the magnetic circuit solution for a single-phase core, which can be adapted to specific core configurations [34].

III. SIMULATION-BASED VERIFICATION OF INDIVIDUAL COMPONENTS USING FEM

This section constructs several simplified winding systems using the commercial FEM software ANSYS, with the objective of verifying the individual components of the proposed method.

A. Simulation-Based Verification for Winding Modeling

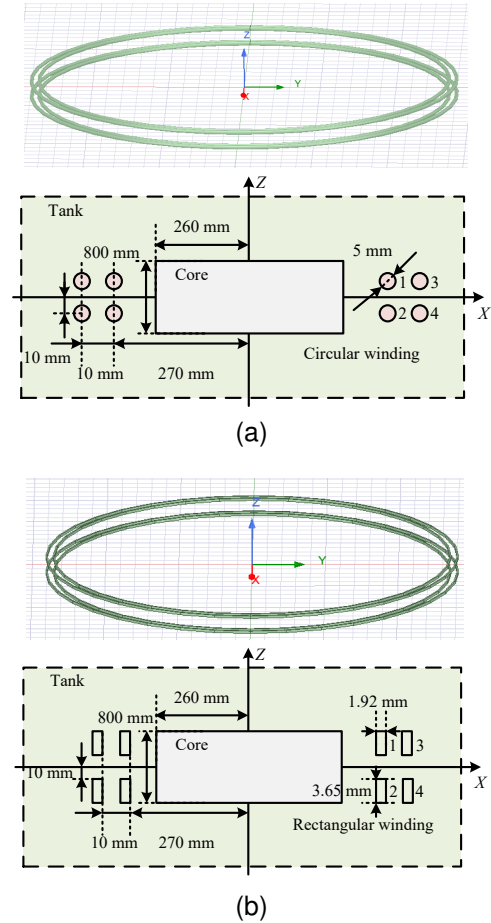


Fig. 6. Winding systems constructed in ANSYS, comprising a 4-turn winding, a core ($R_{core} = 0.26m$, $H_{core} = 0.8m$), and a tank ($0.8m \times 0.8m \times 0.8m$). (a) Winding configuration with a circular cross-section ($R = 2.5mm$); (b) Winding configuration with a rectangular cross-section ($1.92mm \times 3.65mm$).

Eqs. (15)–(18) define the coupled circuit–magnetic circuit

To verify the calculation of the inductance matrix $\mathbf{L}$ and the capacitance matrix $\mathbf{C}$ using the PEEC method, two winding

systems are constructed and analyzed using the magnetostatic and electrostatic solvers in ANSYS, as shown in Fig. 6. Notably, $\mathbf{C}$ denotes the inverse Maxwell capacitance matrix with negative off-diagonal elements. The corresponding results are summarized in Table I, with the subscripts of $\mathbf{L}$ and $\mathbf{C}$ indexed according to the winding numbering shown in Fig. 6. In Table I, PEEC+Loop denotes the PEEC calculation using the same unconnected turn/disk configuration as the FEM model in Fig. 6, where different turns/disks are treated as separate conductors for parameter extraction. By contrast, Fig. 7 shows the practical winding connections corresponding to Fig. 6a, where the turns are physically connected. The corresponding results are denoted as PEEC+Partial in Table I.

As shown in Table I, the inductance matrix $\mathbf{L}$ and the capacitance matrix $\mathbf{C}$ calculated using the PEEC method for both circular and rectangular cross-section windings exhibit close agreement with the FEM results. Specifically, the maximum deviations for the $\mathbf{L}$ and $\mathbf{C}$ matrix elements are within 1% and 9%, respectively. The relatively larger deviations observed in the $\mathbf{C}$ matrix are attributed to the filament approximations in this study [25], where discretization is only performed along the conductor direction without discretizing the cross-section. Furthermore, comparing the PEEC+Loop and PEEC+Partial results indicates that practical winding connections (see Fig. 7) noticeably affect the results, introducing an average deviation of approximately 5%. In terms of computational efficiency, the MATLAB-based PEEC solver yields solutions in merely 0.5 seconds on a workstation equipped with an Intel Ultra 7 265KF CPU and 32 GB of memory. In contrast, ANSYS requires approximately 15 minutes to calculate the corresponding results, demonstrating a speedup ratio of nearly 2000.

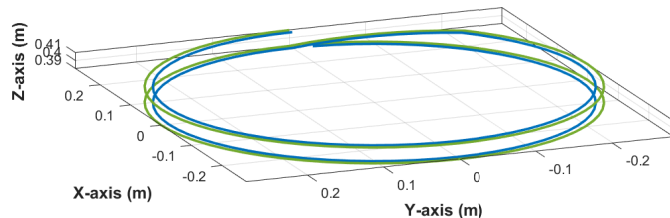


Fig. 7. Practical winding connections corresponding to Fig. 6a, constructed using the PEEC method.

B. Simulation-Based Verification for Core and Tank Boundary Effects

This section examines whether the PEEC method, combined with the image method, can capture the boundary effects of the transformer core and tank on the $\mathbf{L}$ and $\mathbf{C}$ matrices in a manner comparable to FEM. In the ANSYS magnetostatic solver, the core and tank are independently specified as magnetic insulation to calculate $\mathbf{L}$, whereas in the electrostatic field solver, they are independently assigned a potential of 0 V to calculate $\mathbf{C}$. The results for the winding system illustrated in Fig. 6a are summarized in Table II.

As shown in Table II, the maximum deviation in the elements of the inductance matrix $\mathbf{L}$ is within 3%, while that of the capacitance matrix $\mathbf{C}$ is within 5%, thereby confirming

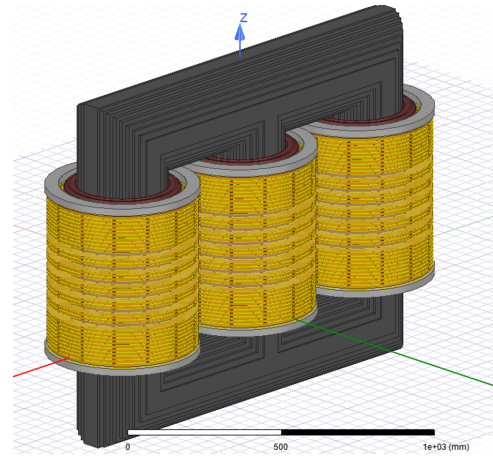


Fig. 8. 3D finite element model of transformer [2]. This study compares PEEC-derived and FEM-derived inductance and capacitance matrices for the HV A-phase winding.

the effectiveness of the image method. Compared with the results in Table I, the presence of the core nearly halves the $\mathbf{L}$ matrix as well as the mutual capacitance of each turn, while markedly increasing the capacitance to ground. The tank exhibits a similar influence, though the corresponding matrix elements decrease by only about one-tenth. Overall, the boundary effect of the core is substantially stronger than that of the tank, consistent with the principle that boundary effects diminish with increasing distance from the boundary. Moreover, even when the tank boundary is represented by 26 image segments for each physical winding segment, the total computation time remains approximately 3 seconds.

In addition to these simple examples, the proposed method is further validated against a complex FEM-based transformer model. Specifically, a comparison is conducted on the HV A-phase of the transformer illustrated in Fig. 8, whose detailed parameters are shown in Ref. [2]. The PEEC-derived inductance and capacitance matrices exhibit average deviations of only 6% and 8% from the FEM results, respectively. In terms of computational efficiency, the FEM simulation requires approximately one day, whereas the PEEC solver completes within 5 minutes, demonstrating a substantial reduction in computation time.

IV. EXPERIMENTAL VALIDATION IN TWO CASE STUDIES

To further validate the proposed method, two practical cases are investigated in this section. The detailed parameters of the corresponding experimental equipment are outlined in Table III.

A. Case Study 1: Specially Designed Winding

As shown in Fig. 9a and Table IV, a specially designed winding is constructed. The copper foil is grounded to provide a controlled conducting/equipotential boundary for validating the image method. It is not intended to reproduce the full behavior of the laminated core. Correspondingly, an experimental platform is constructed with the wiring configuration illustrated in Fig. 9b. A step signal with an amplitude of 100

TABLE I
COMPARISON OF FEM AND PEEC RESULTS FOR THE **L** AND **C** MATRICES UNDER WINDINGS WITH CIRCULAR AND RECTANGULAR CROSS-SECTIONS. INDUCTANCE IS EXPRESSED IN nH, AND CAPACITANCE IN pF.

Inductance Matrix L (nH)											
Cross-section	Method	L_{11}	L_{22}	L_{33}	L_{44}	L_{12}	L_{13}	L_{14}	L_{23}	L_{24}	L_{34}
Circular	FEM	1704.3	1704.3	1780.2	1780.2	911.5	1172.8	896.2	896.2	1172.8	957.9
	PEEC+Loop	1695.2	1695.2	1767.8	1767.8	910.5	1172.0	895.3	895.3	1172.0	956.9
	PEEC+Partial	1811.6	1811.6	1871.3	1871.3	962.9	1236.3	936.9	936.9	1236.3	1009.5
Rectangular	FEM	1856.0	1856.0	1937.1	1937.1	855.6	1171.3	849.1	849.1	1171.3	899.61
	PEEC+Loop	1850.0	1850.0	1930.3	1930.3	853.5	1130.9	842.7	842.7	1130.9	897.28

Capacitance Matrix C (pF)											
Cross-section	Method	C_{11}	C_{22}	C_{33}	C_{44}	C_{12}	C_{13}	C_{14}	C_{23}	C_{24}	C_{34}
Circular	FEM	45.3	45.3	46.7	46.7	-9.5	-26.6	-5.0	-5.0	-26.6	-10.0
	PEEC+Loop	41.2	41.2	42.6	42.6	-9.2	-23.6	-4.5	-4.5	-23.6	-9.6
	PEEC+Partial	42.4	42.4	44.0	44.0	-10.4	-23.8	-4.2	-4.2	-23.8	-11.0
Rectangular	FEM	34.0	34.0	35.1	35.1	-6.6	-18.8	-4.5	-4.5	-18.8	-6.9
	PEEC+Loop	30.1	30.1	31.3	31.3	-6.4	-15.8	-4.2	-4.2	-15.8	-6.7

TABLE II
COMPARISON OF FEM AND PEEC RESULTS FOR THE **L** AND **C** MATRICES UNDER THE WINDING WITH CIRCULAR CROSS-SECTION, CONSIDERING CORE AND TANK BOUNDARY EFFECTS INDIVIDUALLY. INDUCTANCE IS EXPRESSED IN nH, AND CAPACITANCE IN pF.

Inductance Matrix L (nH)											
Case	Method	L_{11}	L_{22}	L_{33}	L_{44}	L_{12}	L_{13}	L_{14}	L_{23}	L_{24}	L_{34}
With core	FEM	793.7	793.7	1064.8	1064.8	166.9	378.0	163.7	163.7	378.0	280.5
	PEEC+Loop	788.5	788.5	1092.0	1092.0	181.7	392.6	174.2	174.2	392.6	283.3
With tank	FEM	1592.3	1592.3	1648.4	1648.4	799.2	1050.7	774.5	774.5	1050.8	825.9
	PEEC+Loop	1572.5	1572.5	1624.0	1624.0	788.2	1040.0	763.8	763.8	1040.0	813.4

Capacitance Matrix C (pF)											
Case	Method	C_{11}	C_{22}	C_{33}	C_{44}	C_{12}	C_{13}	C_{14}	C_{23}	C_{24}	C_{34}
With core	FEM	58.9	58.9	49.0	49.0	-3.4	-22.7	-2.6	-2.6	-22.7	-8.5
	PEEC+Loop	58.7	58.7	47.1	47.1	-3.2	-23.2	-2.5	-2.5	-23.2	-8.5
With tank	FEM	46.1	46.1	47.9	47.9	-9.1	-26.2	-4.4	-4.4	-26.2	-9.2
	PEEC+Loop	44.2	44.2	46.1	46.1	-8.6	-25.8	-3.4	-3.4	-25.8	-8.8

TABLE III
SPECIFICATIONS OF THE EXPERIMENTAL EQUIPMENT

Equipment	Model
Oscilloscope	Tektronix MDO3054
Nanosecond pulse generator	SJTU-1000
Lightning impulse generator	SJTU-500
Voltage probe	Tektronix P5100A

TABLE IV
KEY PARAMETERS FOR THE SPECIALLY DESIGNED WINDING

Parameter	LV Winding	HV Winding
Winding type	Continuous	Continuous
Winding radius (m)	0.28	0.26
Conductor shape	Rectangular	Rectangular
Turn width (m)	1.92×10^{-3}	1.92×10^{-3}
Turn height (m)	3.65×10^{-3}	3.65×10^{-3}
Total winding height (m)	0.388	0.773
Number of disks	100	200
Number of turns per disk	1	1

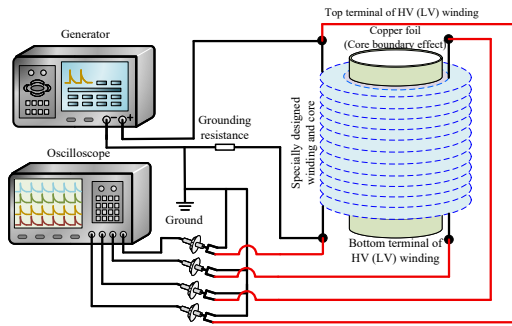
V and a rise time of 5 μ s is injected into the top terminal of the HV winding using a generator, while voltage signals are measured simultaneously at the bottom terminal of the HV winding and at both the top and bottom terminals of the LV winding.

The transient simulations obtained using the proposed method, together with the experimental measurements, are presented in Fig. 10. Specifically, Fig. 10a shows the signal injected at the top terminal of the HV winding. As shown in Figs. 10b-10d, the constructed equivalent circuit effectively captures the physical characteristics of the HV and LV windings, with the rising and falling edges of the measured

signals at each terminal closely aligned with the simulated results. Moreover, a comparison with transient simulations that neglect the core boundary effect reveals that accounting for this effect markedly reduces the coupling between the HV and LV windings, thereby lowering the voltage response in the LV winding. This observation is consistent with the principle that the core boundary effect diminishes the mutual inductance between the HV and LV windings. Additionally, the computation of the PEEC-related matrices for a 300-turn



(a)



(b)

Fig. 9. Experimental measurement setup for the specially designed winding. The copper foil is grounded to emulate the boundary condition. (a) Actual wiring configuration. (b) Measurement wiring configuration.

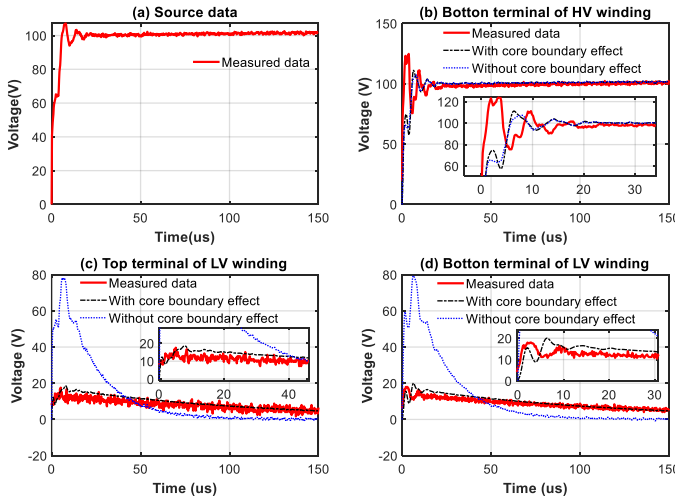


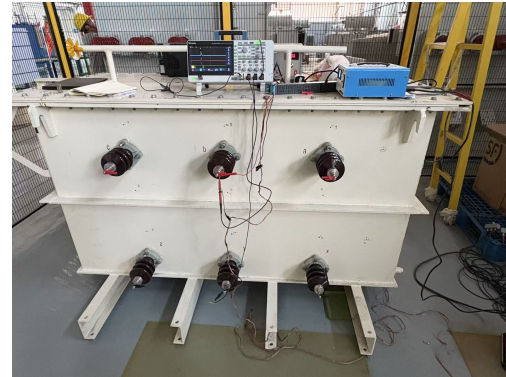
Fig. 10. Comparison between measured and simulated data of the specially designed winding at different observation points. (a) Injected signal. (b) Bottom terminal of the HV winding. (c) Top terminal of the LV winding. (d) Bottom terminal of the LV winding.

winding system required only 21 seconds. For a transient simulation of 150 μ s with a step size of 0.2 μ s, the computation is completed in less than 5 seconds.

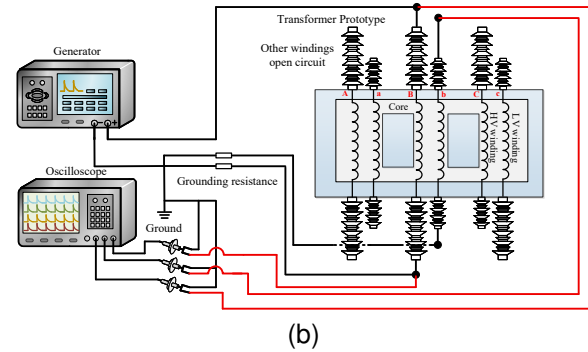
B. Case Study 2: Transformer Prototype

As shown in Fig. 11a, an oil-immersed power transformer prototype is manufactured, with its key parameters listed in

Table V. The prototype is based on a 110 kV power transformer design with one-third-scale physical geometry. The transformer employs a three-phase double-winding configuration with a 3/2 iron core, and the HV and LV windings of phase B comprise 44 continuous winding disks, each consisting of 9 turns. Correspondingly, an experimental platform is constructed with the wiring configuration illustrated in Fig. 11b. An 183 V lightning impulse with a rise time of 0.96 μ s and a tail time of 49.28 μ s is injected into the top terminal of the HV winding using a generator, while voltage signals are measured simultaneously at the bottom terminal of the HV winding and at the top terminals of the LV winding.



(a)



(b)

Fig. 11. Experimental measurement diagram for the transformer prototype. (a) Actual wiring diagram. (b) Measurement wiring diagram.

TABLE V
KEY PARAMETERS FOR TRANSFORMER PROTOTYPE

Parameter	LV Winding	HV Winding
Winding type	Continuous	Continuous
Winding radius (m)	0.1815	0.1335
Conductor shape	Rectangular	Rectangular
Turn width (m)	2.5×10^{-3}	2.5×10^{-3}
Turn height (m)	8.3×10^{-3}	8.3×10^{-3}
Total winding height (m)	0.505	0.505
Number of disks	44	44
Number of turns per disk	9	9

Fig. 12 presents the transient simulations obtained with the proposed method for the B-phase HV and LV windings, together with the corresponding experimental measurements. The injected signal at the HV winding terminal is shown in Fig. 12a. As illustrated in Figs. 12b–12c, the rising and falling edges in both windings are well captured, although subsequent

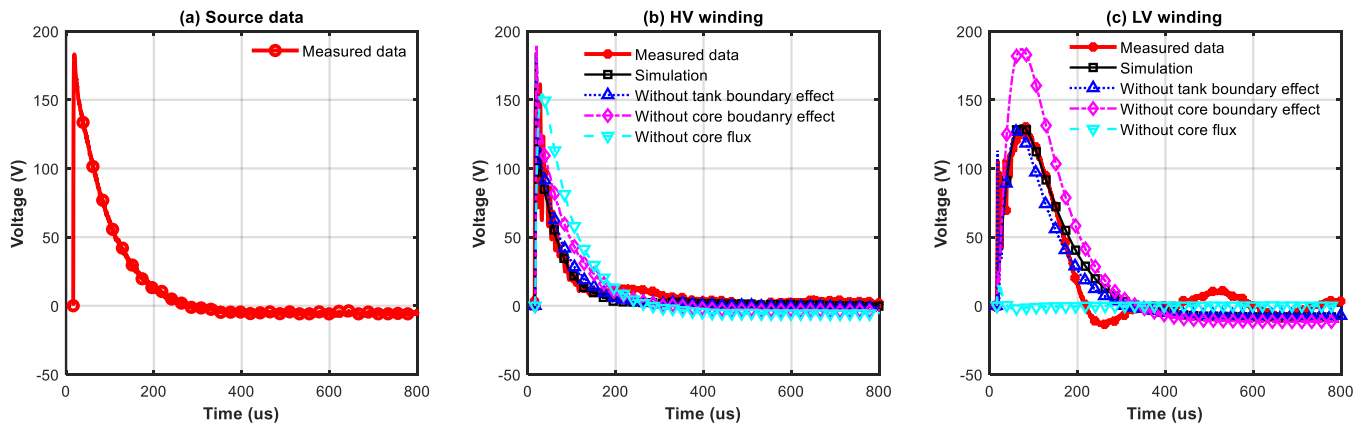


Fig. 12. Comparison between measured and simulated data of the transformer prototype at different observation points. (a) Injected signal. (b) Bottom terminal of the HV winding. (c) Top terminal of the LV winding.

oscillations in the LV winding, particularly at the later stage of the transient response, are not accurately reproduced and are likely attributable to measurement effects. Furthermore, the boundary effect of the transformer tank is considerably weaker than that of the core, primarily because the tank is located farther from the winding. A comparison with Case 1 in Fig. 10 further indicates that the boundary effect of the core is weaker in Case 2, owing to two factors: the winding is positioned farther from the core, and each disk comprises multiple turns, with only the inner turns being significantly affected. In addition, the magnetic circuit exhibits a frequency-dependent low-pass behavior with a cutoff frequency of 4.3 kHz. For signals with higher frequency content (e.g., very fast pulses), flux penetration into the core is strongly suppressed by eddy currents, so the coupling occurs primarily through the outside flux. In contrast, low-frequency components penetrate the core more effectively and are coupled from the HV winding to the LV winding with less attenuation. As shown in Fig. 12c, the waveform exhibits a markedly slower rising edge compared with the injected signal in Fig. 12a. This phenomenon is observed in the prototype with a core but is absent in Case 1 without a core. Additionally, the computation of the PEEC-related matrices for a 792-turn winding system required only 120 seconds. For a transient simulation of 800 μ s with a step size of 0.2 μ s, the computation is completed in less than 5 seconds.

V. DISCUSSION

This section further examines the sensitivity of the image method, assesses the applicability of the proposed method to different winding configurations, compares it with existing numerical methods, and discusses its remaining limitations.

A. Sensitivity Analysis for Core and Tank Image Assumptions

The proposed method uses corresponding image segments to approximately account for the boundary effects of the transformer core and tank. Since their accuracy depends on the boundary geometry and the clearance between the winding and the boundary, a sensitivity analysis is performed in this subsection. The overall deviation is quantified using the relative error

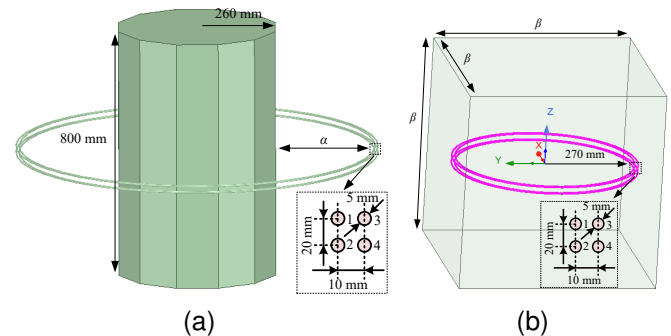


Fig. 13. Constructed winding systems for analyzing the image assumptions of the core and tank. (a) Winding system for the core image assumption (α denotes the distance between the winding and core). (b) Winding system for the tank image assumption (β denotes the height, length, and width of the tank).

between PEEC and FEM results. The constructed winding systems used for this analysis are illustrated in Fig. 13.

The spherical image method provides a localized approximation rather than an exact analytical solution for the cylindrical core boundary. This approach remains effective because the corrections to the self parameters and the mutual inductive and capacitive couplings between nearby segments depend mainly on local boundary proximity and curvature, which are closely matched by the equivalent spherical surface near the winding. In contrast, the mutual inductive and capacitive couplings between more distant segments are captured less accurately because they depend more strongly on the global boundary geometry. However, their contribution to the total matrix error is limited because their magnitudes are generally smaller than those of the self terms and near-neighboring mutual terms. Sensitivity analysis validates this mechanism in Figs. 14a-14b. Varying the winding-to-core distance α from 20 mm to 240 mm maintains the inductance error between 5.37% and 6.99%, while simultaneously reducing the capacitance error from 9.73% to 3.9%. As the winding moves farther from the core, the inherent attenuation of the boundary effect overrides the increasing geometric discrepancy between the cylindrical and spherical profiles, thereby suppressing the total matrix

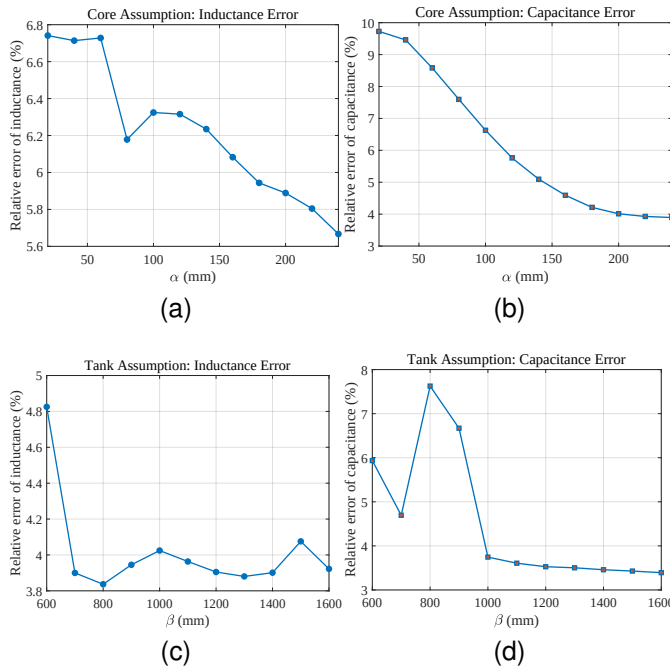


Fig. 14. Comparison of PEEC-image and FEM results under different settings. (a) Relative inductance error for the constructed winding system in Fig. 13a. (b) Relative capacitance error for the constructed winding system in Fig. 13a. (c) Relative inductance error for the constructed winding system in Fig. 13b. (d) Relative capacitance error for the constructed winding system in Fig. 13b.

error at larger distances.

The cuboid tank is modeled using 26 localized image segments rather than six infinite planes to capture the primary tank boundary effect. Higher-order image segments are excluded since the substantial clearance between the winding and the tank renders their influence negligible. As demonstrated by the results in Figs. 14c to 14d, varying winding-to-tank distance β from 600 mm to 1600 mm maintains the inductance matrix error between 3.84% and 4.83% and reduces the capacitance matrix error from 7.63% to 3.39%. These data indicate that the tank boundary effect weakens significantly with the increase in winding-to-tank clearance.

B. Modeling of Different Winding Configurations

The PEEC-based method extracts parameters at the segment level and assembles them according to practical winding connections, enabling different winding configurations to be represented. After conductor discretization, the same $\mathbf{L}$, $\mathbf{P}$, and $\mathbf{R}$ extraction procedure can be combined with appropriate incidence matrices or node-merging operations to model continuous, disk-type, and interleaved windings [22]. To evaluate this topological adaptability, Fig. 15 compares last-disk voltage responses of continuous and interleaved windings, both of which share the key parameters listed in Table VI but differ in winding type, under a 1.2/50 μs lightning impulse. Under identical excitation, the continuous winding peaks at 19.4 μs , whereas the interleaved variant peaks at 78.0 μs , demonstrating that the proposed method effectively reflects internal transient distribution variations induced by diverse winding configurations.

TABLE VI
KEY PARAMETERS FOR CONTINUOUS AND INTERLEAVED WINDING CASES

Parameter	LV Winding	HV Winding
Winding radius (m)	0.30	0.23
Conductor shape	Rectangular	Rectangular
Turn width (m)	1.92×10^{-3}	1.92×10^{-3}
Turn height (m)	3.65×10^{-3}	3.65×10^{-3}
Total winding height (m)	0.4	0.4
Number of disks	8	8
Number of turns per disk	4	4

C. Comparison with Other Numerical Methods

The proposed method should be regarded as a complementary white-box modeling approach [38]. As summarized in Table VII, conventional transformer modeling methodologies, including FEM-based methods [2], [20], [21], [23], multi-conductor transmission-line (MTL) formulations [30], [36], [37], black-box models, and gray-box models, possess distinct operational features and application domains.

In contrast to these established paradigms, the primary advantage of the proposed framework becomes particularly evident in applications necessitating intensive repetitive evaluations, such as parametric sweeps, winding structure optimizations, insulation design iterations, uncertainty analyses, and fault diagnosis studies [1], [2], [39]. For these scenarios, although the computational cost of an individual FEM simulation remains manageable, thousands of FEM simulations are often computationally prohibitive. The proposed framework addresses this bottleneck by rapidly screening numerous candidates, ensuring that only a minimal subset of selected cases requires subsequent verification through targeted FEM simulations or experimental testing. Furthermore, once nodal voltages are resolved, the local voltage gradient, defined as $\max |dv_n(t)/dt|$, is evaluated for each monitored disk or turn to characterize transient severity. This formulation enables a comparison of transient behavior, including voltage magnitude, rise time, peak time, and internal dv/dt profiles. Unlike black-box or gray-box models that are restricted to terminal behavior fitting, the proposed PEEC approach maps internal physical variables directly from transformer structures. This predictive capability provides vital insights into internal insulation design and transient response evaluation.

D. Limitations and Future Prospects

The proposed PEEC-based framework is intended for fast transient studies with physically interpretable circuit parameters and internal voltage-distribution calculation. Nevertheless, several limitations should be noted.

First, the present formulation does not explicitly include frequency-dependent conductor, dielectric, and magnetic properties. Skin and proximity effects may modify the effective resistance and inductance of winding conductors, thereby affecting damping, resonant frequencies, and transient overvoltage prediction. In addition, high-frequency dielectric losses and insulation leakage are treated in a simplified manner [2], [20], [23], [40]. Future work will incorporate frequency-dependent

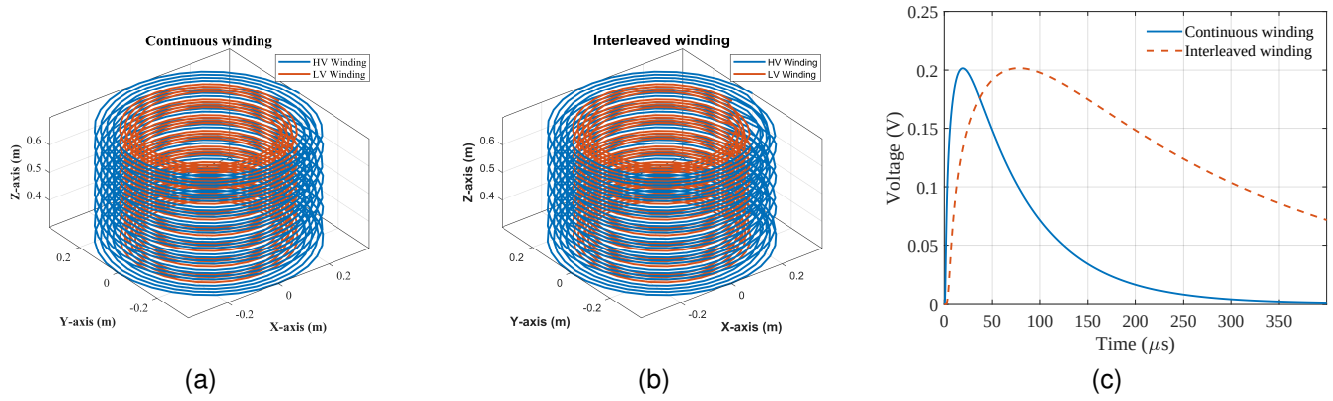


Fig. 15. Illustration and transient-voltage comparison of different winding configurations. (a) Continuous winding. (b) Interleaved winding. (c) Calculated voltage response at the last disk of the HV winding under a 1.2/50 μ s lightning impulse.

TABLE VII
COMPARISON OF THE PROPOSED METHOD WITH REPRESENTATIVE TRANSFORMER MODELING METHODS

Method	Modeling Principle	Advantages	Limitations
Analytical lumped-parameter model [8], [10], [11], [24], [29]	The winding is represented by an RLC ladder network, with parameters derived from simplified geometrical formulations.	Simple implementation; clear physical and circuit interpretation; low computational cost.	Accuracy strongly depends on simplifying assumptions; detailed electromagnetic coupling, high-frequency effects, core/tank boundary effects, and multi-winding interactions are difficult to represent.
FEM-based model [2], [20], [21], [23]	RLC parameters are extracted from 2-D/3-D field solutions and assembled into an equivalent circuit. FEM-BEM formulations can further improve open-boundary treatment.	Higher parameter accuracy than analytical models; capable of representing material properties, skin/proximity effects, and complex field distributions.	Requires meshing and repeated field solutions; high-fidelity 3-D models are computationally intensive and may require geometrical or physical simplifications.
Distributed-parameter / transmission-line model [36], [37]	The winding is formulated as a multi-conductor transmission-line system.	Effective for analyzing high-frequency wave propagation and winding resonances.	May involve large coupled matrices and numerical stability issues; practical connections, core/tank effects, and transformer-level structural details are difficult to incorporate.
Black-box / gray-box model [12]–[14], [17]–[19]	Terminal frequency responses are fitted using rational functions or equivalent circuits.	Efficient for electromagnetic transient simulations; suitable when only terminal behavior is required.	Limited physical interpretability; internal voltage distribution and local insulation stress are difficult to obtain; geometry or topology changes require re-identification.
Proposed PEEC-based method	The winding is discretized into PEEC segments; core/tank boundary effects are approximated by image segments; core flux is represented by a magnetic circuit.	Circuit-oriented and physically interpretable; avoids repeated full-domain field solutions; accommodates practical winding connections; enables internal transient-voltage calculation.	Core/tank boundary effects are approximated; frequency-dependent conductor characteristics and nonlinear magnetic behavior require further extension.

conductor parameters, dielectric-loss branches, and vector-fitting techniques to extend the model toward frequency-response analysis [41]–[43].

Second, the image method provides an efficient approximation of core and tank boundary effects. The spherical core image and cuboid tank images are suitable for the cases considered in this study, as supported by the FEM comparisons and sensitivity analysis. However, strong asymmetry, very small clearances, irregular tank shapes, or local protrusions may require higher-order image representations or hybrid PEEC-FEM formulations.

Finally, the experimental validation is limited to a specially designed winding and a scaled transformer prototype. Future work should consider larger three-phase transformers, additional winding configurations, more field data, and broader operating conditions. Ultimately, this framework serves as an efficient tool for screening and design exploration to complement, rather than replace, high-fidelity FEM simulations and physical testing.

VI. CONCLUSION

To reduce the computational cost and time associated with white-box transformer models while accounting for practical winding connections, this study proposes a fast electromagnetic transient modeling framework for transformers using the PEEC and image methods. Based on both simulation and experimental results, the following conclusions can be drawn:

- 1) The PEEC-derived $\mathbf{L}$ and $\mathbf{C}$ matrices of the winding are broadly consistent with those obtained from FEM-based approaches, while requiring significantly less computation time. Moreover, practical winding connections influence the calculation of the relevant parameters.
- 2) By integrating the image method with PEEC, the $\mathbf{L}$ and $\mathbf{C}$ matrices can be adjusted through image segments, thereby imposing boundary conditions comparable to those in FEM. In general, the boundary effect of the core is considerably stronger than that of the tank due to their locations, and both effects reduce the associated inductive and capacitive couplings.

- 3) The coupled circuit-magnetic circuit exhibits a low-pass behavior, attenuating high-frequency components of the injected signal while facilitating the coupling of low-frequency components from one winding to another.

APPENDIX A

DEFINITION OF CORE RELUCTANCE AND EDDY-CURRENT INDUCTANCE

This appendix gives the definitions of the core reluctance R_m and the equivalent eddy-current inductance G_e used in the coupled circuit-magnetic-circuit model [33], [34]. For a magnetic branch with mean magnetic path length l , effective cross-sectional area A , core flux ϕ , and average flux density $B = \phi/A$, the magnetomotive force (MMF) is expressed as:

$$U_m = Hl \quad (A1)$$

where H is the magnetic field intensity. It should be noted that the contribution of leakage flux to the MMF has not been excluded for the sake of convenience of derivation. If the eddy-current effect is not considered, the static magnetic characteristic gives:

$$U_h = H_h l \quad (A2)$$

where H_h is determined by the static B - H curve. The corresponding core reluctance is defined as:

$$R_m = \frac{U_h}{\phi} \quad (A3)$$

For a linear magnetic branch, this definition reduces to:

$$R_m = \frac{l}{\mu A} \quad (A4)$$

where μ is the permeability of the core material. For nonlinear magnetization, R_m can be evaluated from the static magnetization curve as a secant or incremental reluctance.

G_e accounts for the eddy-current shielding effect in the laminated core during rapid flux variation. Based on the one-dimensional field-diffusion approximation for a thin lamination with thickness δ , permeability μ , and conductivity σ , the magnetic field penetration can be described by:

$$\frac{\partial^2 H(x, t)}{\partial x^2} = \mu\sigma \frac{\partial H(x, t)}{\partial t} \quad (A5)$$

For a step excitation at the lamination surface, the average flux density can be approximated by a first-order relation:

$$H = H_h + a \frac{dB}{dt} \quad (A6)$$

where

$$a = \frac{1}{3.23} \frac{\sigma \delta^2}{4} \quad (A7)$$

Multiplying by the magnetic path length l , and using $\phi = BA$, yields:

$$U_m = U_h + G_e \frac{d\phi}{dt} \quad (A8)$$

where

$$G_e = \frac{al}{A} = \frac{1}{3.23} \frac{\sigma \delta^2 l}{4A} \quad (A9)$$

Thus, G_e is determined by the lamination conductivity, lamination thickness, magnetic path length, and effective core

area. It represents the additional MMF component associated with eddy-current damping during fast flux variation.

When the static magnetic branch is linearized as $U_h = R_m \phi$, the magnetic branch equation becomes:

$$U_m = R_m \phi + G_e \frac{d\phi}{dt} \quad (A10)$$

Accordingly, the magnetic branch has a first-order time constant:

$$\tau_e = \frac{G_e}{R_m} \quad (A11)$$

The corresponding cutoff frequency can be estimated as:

$$f_c = \frac{R_m}{2\pi G_e} \quad (A12)$$

Below this characteristic frequency, the core-flux branch can influence the transient response. Above this frequency, the core-flux contribution is attenuated, and the response is mainly governed by the winding capacitance, leakage-field coupling, and boundary effects. For a multi-branch magnetic circuit, R_m and G_e are calculated for each branch as:

$$R_{m,j} = \frac{U_{h,j}}{\phi_j} \quad G_{e,j} = \frac{1}{3.23} \frac{\sigma_j \delta_j^2 l_j}{4A_j} \quad (A13)$$

where l_j , A_j , σ_j , and δ_j are the mean magnetic path length, effective cross-sectional area, lamination conductivity, and lamination thickness of the j th magnetic branch, respectively.

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